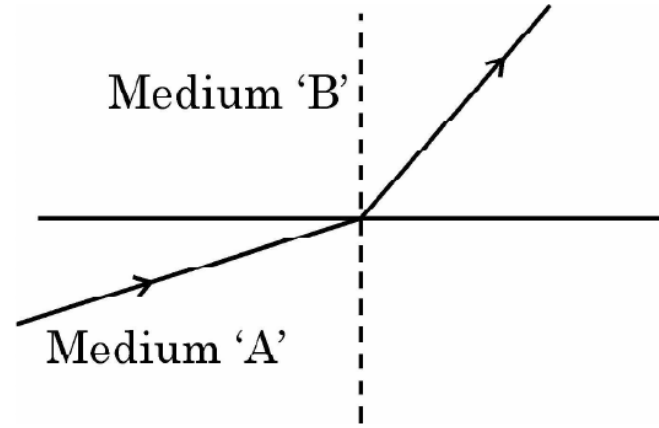


- (b) (i) Define absolute refractive index.
- (ii) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find :
- (I) Speed of light in vacuum
- (II) Speed of light in water

23. A light ray enters from medium A to medium B as shown in the figure.

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- (a) Which one of the two media is denser w.r.t. other medium ? Justify your answer. 1
- (b) If the speed of light in medium A is v_a and in medium B is v_b , what is the refractive index of B with respect to A. 1

OR

- (a) A ray of light starting from diamond is incident on the interface separating diamond and water. Draw a labelled ray diagram to show the refraction of light in this case. 1
- (b) Absolute refractive indices of diamond and water are 2.42 and 1.33 respectively. Find the value of refractive index of water w.r.t. diamond. 1

38. The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the large refractive index is optically denser medium than the other. The other medium with a lower refractive index is optically rarer. Also the speed of light through a given medium is inversely proportional to its optical density.

- (i) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is 3×10^8 m/s.
- (ii) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say θ), then write the increasing order of the angle of refraction in these media.
- (iii) (A) The speed of light in glass is 2×10^8 m/s and in water is 2.25×10^8 m/s.
 - (a) Which one of the two is optically denser and why ?
 - (b) A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass ? Give reason.

OR

CBSE 23

- (iii) (B) The absolute refractive indices of water and glass are $4/3$ and $3/2$ respectively. If the speed of light in glass is 2×10^8 m/s, find the speed of light in (i) vacuum and (ii) water.

37. When a ray of light moving in a medium enters obliquely in another medium, it bends from its path. This phenomenon is called refraction of light. The ability of a medium to refract light is expressed in terms of optical density. The term optical density is not the same as mass density. If we compare two optical media, the one with the larger absolute refractive index is an optically denser medium than the other. Also, the speed of light through a given medium is inversely proportional to its optical density.

- (a) Define the term 'absolute refractive' index of a medium. 1
- (b) Refractive indices of glass, water and carbon disulphide are 1.50, 1.33 and 1.62 respectively. In which one of these media is the speed of light (i) highest, and (ii) lowest ? 1
- (c) (i) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in water is 2.25×10^8 m/s, find the speed of light in (I) vacuum, and (II) glass. 2

CBSE 23C

OR

- (ii) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. Find the refractive index of water with respect to glass. 2

23. A ray of light falls making an angle of incidence θ on the surface of a glass slab. Draw a labelled ray diagram to show its path. Also mark lateral displacement on it.

15. The speed of light in vacuum is 3×10^8 m/s. If the speed of light in a medium is 2.25×10^8 m/s, the absolute refractive index of the medium is : **1**

(A) $\frac{7}{6}$

(B) $\frac{5}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

24. (a) State two laws of refraction of light.

2

OR

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- (b) Define the term absolute refractive index of a medium. A ray of light enters from vacuum to glass of absolute refractive index 1.5. Find the speed of light in glass. The speed of light in vacuum is 3×10^8 m/s.

2

14. Absolute refractive index of glass and water is $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4} \times 10^8$ m/s

(B) $\frac{5}{2} \times 10^8$ m/s

CBSE 24

(C) $\frac{7}{3} \times 10^8$ m/s

(D) $\frac{16}{9} \times 10^8$ m/s

- (b) (i) State laws of refraction of light.
- (ii) Draw a ray diagram to show refraction of a ray of light through a rectangular glass slab. How is the emergent ray related to incident ray ? Mark lateral displacement in the diagram.

1. The refractive index of medium 'X' with respect to medium 'Y' is $\frac{2}{3}$ and refractive index of medium 'Y' with respect to medium 'Z' is $\frac{4}{3}$. Find the refractive index of medium 'Z' with respect to medium 'X'.

2. The refractive index of water with respect to air is $\frac{4}{3}$. What is the refractive index of air with respect to water?

3. The refractive index of glass with respect to air is 1.65 and that of water w.r.t air is 1.33. Calculate refractive index of water w.r.t to glass.

4. In an experiment with a glass slab, a student observed that a ray of light incident at an angle of 60° with the normal on one face of the slab, after refraction, strikes the opposite face of the slab before emerging out in air making an angle of 42° with the normal. Draw a labelled diagram to show the path of this ray. What would be the value of angle of refraction and angle of emergence?

5. Refractive index of water and benzene w.r.t air are 1.33 and 1.50 respectively. Calculate refractive index of benzene w.r.t water?

6. The absolute refractive index of 2 media 'A' and 'B' are 2.0 and 1.5 respectively. If the speed of light in medium 'B' is 2×10^8 m/s, calculate the speed of light in: (i) vacuum
(ii) medium 'A'.

7. The absolute refractive index of glass and water are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, calculate the speed of light in: (i) vacuum (ii) water.

8. If the angle of incidence (i) for a light ray in air be 45° and the angle of refraction (r) in glass be 30° , find the refractive index of glass w.r.t air.

9. Refractive index of air w.r.t is 1.33. What is the value of refractive index of air w.r.t water?

Q.2. Light travels from rarer medium 1 to a denser medium 2. The angle of incident and refraction are respectively 45° and 30° . Calculate the (i) refractive index of second medium with respect to the first medium and (ii) refractive index of medium 1 with respect to the medium 2.

Q.4. How much time will light take to cross 2 mm thick glass pane if refractive index of glasses is $\frac{3}{2}$?

Q.6. A ray of light passes from air to glass ($n = 1.5$) at an angle of 30° . Calculate the angle of refraction.

Q.8. The refractive index of diamond is 2.47 and that of glass is 1.51. How much faster does light travel in glass than in diamond?

Q.11. A ray of light is traveling from glass to air. The angle of incidence in glass is 30° and angle of refraction in air is 60° . What is the refractive index of glass w.r.t air?

1. The refractive index of a material depends upon

(a) Temperature (b) Wavelength of light

(c) Nature of the material (d) All the above

2. When light travels from glass to water separated by a sharp boundary then it

(a) proceeds undeviated (b) bends towards the normal

(c) bends away from the normal (d) is reflected back

4. Velocity of light is maximum in

(a) diamond

(b) water

(c) glass

(d) vacuum

**Q : An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light.
If the emergent beam gets displaced laterally, the device 'X' is :**

(a) Plane mirror

(b) Convex lens

(c) Glass slab

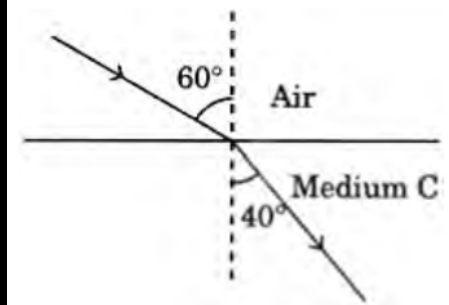
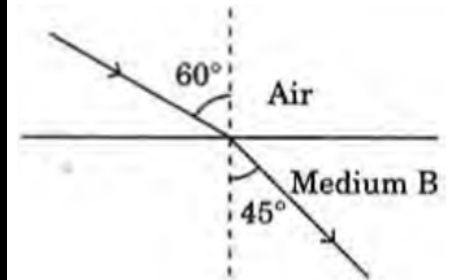
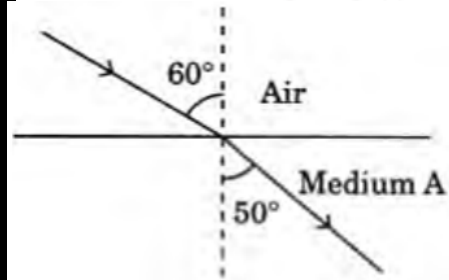
(d) Glass Prism

CBSE 2025



(a) Define absolute refractive index.

(b) The path of a light ray from three different media A, B and C for a given angle of incidence is shown below. Study the diagrams and answer the following questions:



- (i) Which of the three media A, B or C has maximum optical density?
- (ii) Through which of the three media, will the speed of light be maximum?
- (iii) Will the light travelling from A to B bend towards or away from the normal?
- (iv) Will the refractive index of B relative to C be more than unity or less than unity?

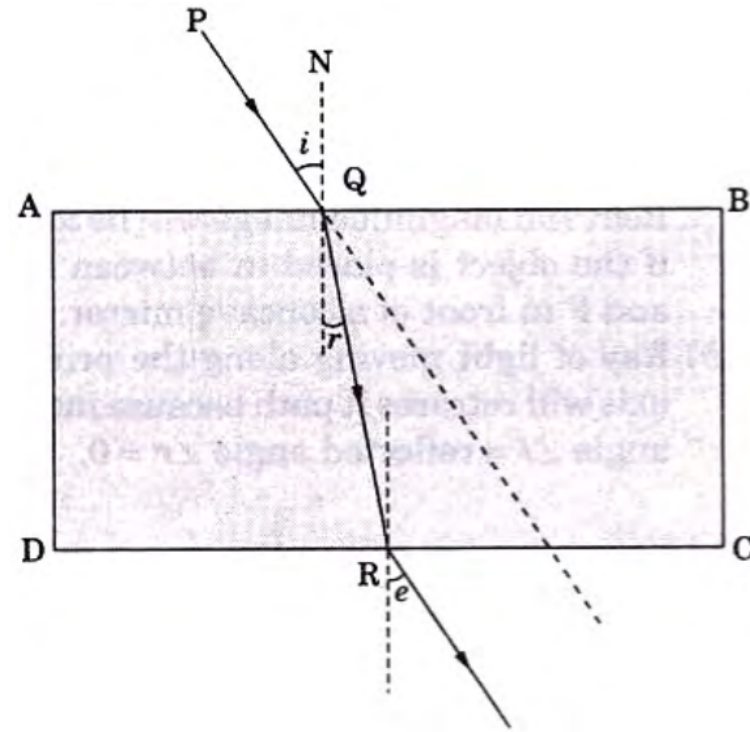
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[All India 2015]

(c) In an experiment to study refraction through a glass slab, it is observed that a ray of light undergoing refraction emerges parallel to the direction of incident ray. Why does it happen so? Explain with the help of a diagram.

Ans :

[All India 2011]



- c. The opposite faces of rectangular glass slab AB and DC are parallel and extent of bending on opposite faces is equal and opposite $\angle i = \angle e$. Hence incident ray is parallel to emergent ray.

- ✕✎ (a) State Snell's law of refraction.
- (b) When a ray of light travelling in air enters obliquely into a glass slab, it is observed that the light ray emerges parallel to the incident ray but it is shifted sideways slightly. Draw a ray diagram to illustrate it.

Ans :

[All India 2014]



- (a) A ray of light falls normally on a face of a glass slab. What are the values of angle of incidence and angle of refraction of this ray?
- (b) Light enters from air to a medium X. Its speed in medium X becomes $1.5 \times 10^8 \text{ ms}^{-1}$. Find the refractive index of medium X.

Ans :

[All India 2012]

- a. When a ray of light falls normally on a glass slab then $i = 0$, and $r = 0$
- b. $v_m = 1.5 \times 10^8 \text{ m/s}$

$$c = 3 \times 10^8 \text{ m/s}$$

We know Refractive index

$$n = \frac{c}{v_m}$$

$$\text{or } n = \frac{3 \times 10^8}{1.5 \times 10^8} = 2$$

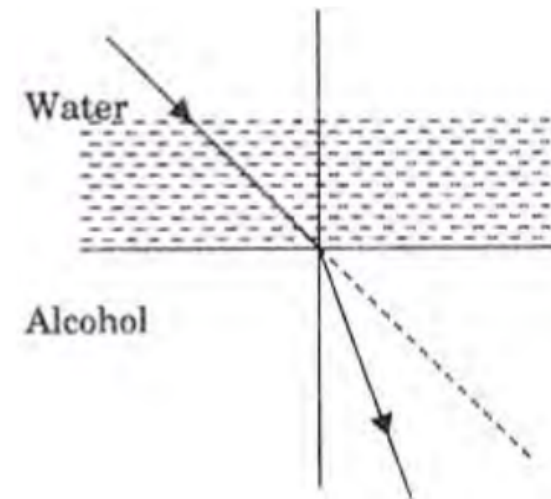
$$n = 2$$

- ✂️ (a) Water has refractive index 1.33 and alcohol has refractive index 1.36. Which of the two medium is optically denser? Give reason for your answer. Draw a ray diagram to show the path of a ray of light passing obliquely from water to alcohol.
- (b) The absolute refractive index of diamond is 2.42 and the absolute refractive index of glass is 1.50. Find the refractive index of diamond with respect to glass.

Ans :

[CBSE 2013]

- a. More refractive index means more optical denser medium. Here refractive index of water is 1.33 and of alcohol is 1.36 $n_{Al} > n_w$. Hence alcohol is more optical denser than water.

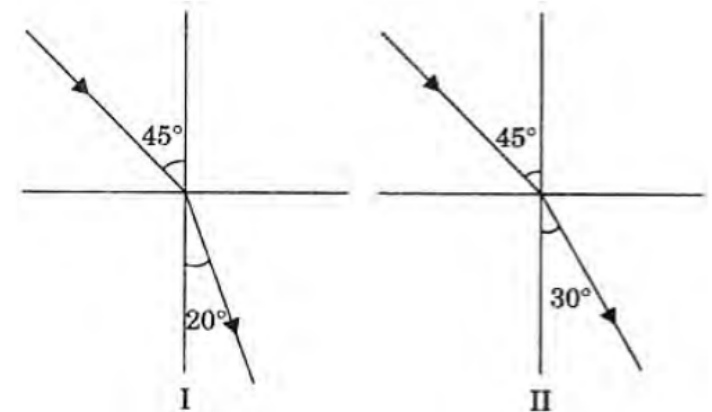


- ✓✍ (a) For the same angle of incidence 45° , the angle of refraction in two transparent media; I and II is 20° and 30° respectively. Out of I and II, which medium is optically denser and why?
- (b) Light enters from air to diamond which has refractive index of 2.42. Calculate the speed of light in diamond, if speed of light in air is $3 \times 10^8 \text{ ms}^{-1}$.

Ans :

[All India 2011]

- a. For the same angle of incidence. The refracted ray in medium I deviates less than in medium II. Hence medium I is more denser than medium II.



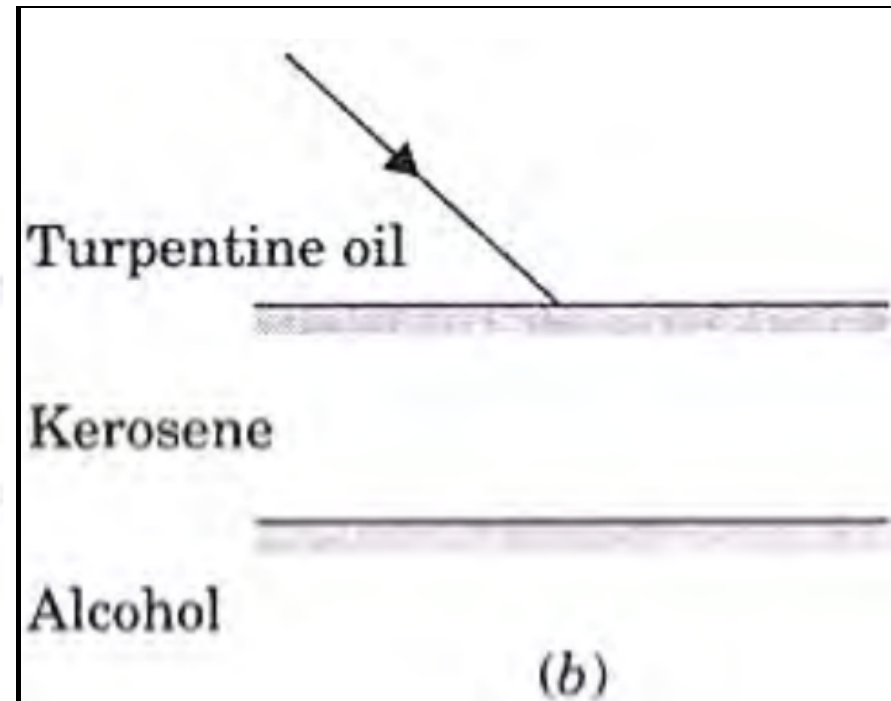
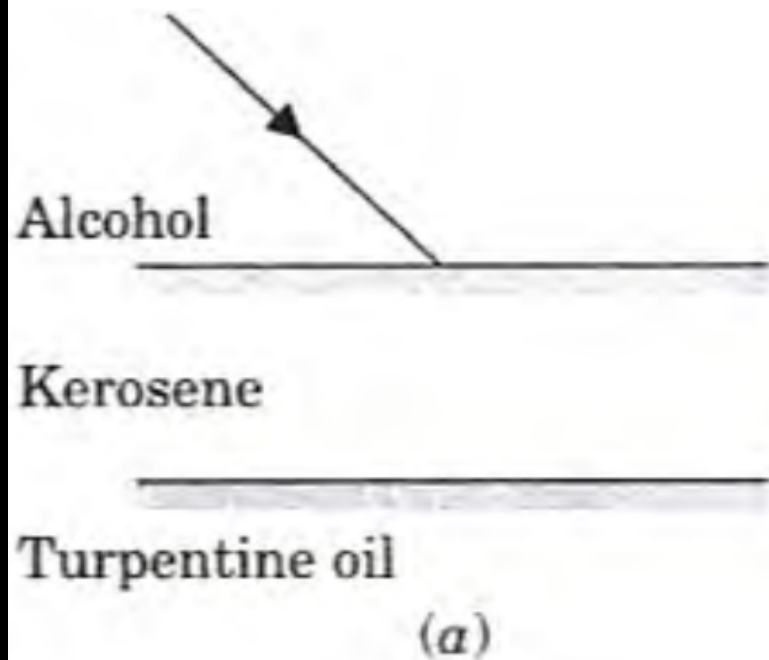
- b. Refractive index of diamond = 2.42
Speed of light in air = $3 \times 10^8 \text{ m/s}$
Speed of light in diamond = ?

$$n = \frac{\text{Speed of light in air}}{\text{Speed of light in diamond}} = \frac{c}{v}$$

$$v = \frac{c}{n} = \frac{3 \times 10^8}{2.42}$$

$$v = 1.24 \times 10^8 \text{ m/s}$$

✓✓✎ Refractive index of turpentine oil, kerosene and alcohol are 1.47, 1.44 and 1.36 respectively. On the basis of this information, complete the following ray diagram to show path of ray of light through each medium. Give reason for your answer.



✂ The refractive index of diamond is 2.42. What is the meaning of this statement?

Ans :

[CBSE 2011]

The meaning of refractive index of diamond is 2.42 is that speed of light in diamond is $1/2.42$ times the speed of light in air/vacuum D,
i.e. speed of light in diamond

$$v = \frac{c}{n} = \frac{3 \times 10^8}{2.42} = 1.24 \times 10^8 \text{ m/s}$$

-  (a) What happens to a ray of light when it travels from one medium to another having equal refractive indices?
- (b) State the cause of refraction of light.

Ans : [CBSE 2011]

- (a) If a ray travels from one medium to another medium of same refractive indices then it will not bend and travels in a straight line.
- (b) Change in speed of light is the cause of refraction.
-